Year 11 Mathematics

Number 8

1.2 Prime Factors

Prime Factors

STARTER · 5 MIN

Date: _____ **Title:** Prime Factors

LEARNING OBJECTIVES

- Recognise prime numbers
- Write a number as a product of prime factors
- Use index notation for prime factors

STARTER *Answer in your book. No calculators.*

1. List the factors of 20
2. List the prime numbers less than 20
3. Is 51 prime? Explain.
4. Work out $2^3 \times 5$

Starter — Answers

STARTER

Tick or correct in green pen

1. **1, 2, 4, 5, 10, 20**

2. **2, 3, 5, 7, 11, 13, 17, 19**

3. $51 = 3 \times 17$, **Not prime**

4. $8 \times 5 =$ **40**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can recognise prime numbers	☐
SILVER	I can write a number as a product of prime factors	☐
GOLD	I can use index notation for prime factors	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Prime Factors

INTRODUCTION · 3 MIN

Every whole number above 1 can be written as a product of prime numbers.

Prime number

Exactly two factors: 1 and itself.

2, 3, 5, 7, 11, 13, …

Factor tree

Split into factors until every branch is prime.

$36 \rightarrow 4 \times 9 \rightarrow 2 \times 2 \times 3 \times 3$

Index form

Use powers for repeated primes.

$36 = 2^2 \times 3^2$

Order

Write the primes from smallest to largest.

$2 \times 3 \times 5$

Examples 1 and 2 — Factor Trees

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Write 24 as a product of prime factors.

Example 2

Write 90 as a product of prime factors.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$24 = 4 \times 6$$
$$= 2 \times 2 \times 2 \times 3$$
$$= \mathbf{2^3 \times 3}$$

Tip: keep splitting until every number is prime.

Example 2

$$90 = 9 \times 10$$
$$= 3 \times 3 \times 2 \times 5$$
$$= \mathbf{2 \times 3^2 \times 5}$$

Tip: write the primes in order.

Task 1

PRACTICE · 9 MIN

BRONZE

Write each number as a product of prime factors

1. 12
2. 30
3. 18
4. 50
5. 42

★ Is 2 the only even prime number? Explain.

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $12 = 2 \times 2 \times 3 = \mathbf{2^2 \times 3}$

2. $30 = \mathbf{2 \times 3 \times 5}$

3. $18 = 2 \times 3 \times 3 = \mathbf{2 \times 3^2}$

4. $50 = 2 \times 5 \times 5 = \mathbf{2 \times 5^2}$

5. $42 = \mathbf{2 \times 3 \times 7}$

★ **Yes, every other even number has 2 as a factor**

Examples 3 and 4 — Index Notation

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

Write 120 as a product of prime factors

in index form.

Example 4

Write 196 as a product of prime factors

in index form.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$120 = 2 \times 2 \times 2 \times 3 \times 5$$

$$= \mathbf{2^3 \times 3 \times 5}$$

Tip: count each repeated prime.

Example 4

$$196 = 4 \times 49$$

$$= 2 \times 2 \times 7 \times 7$$

$$= \mathbf{2^2 \times 7^2}$$

Tip: 196 is a square number.

Task 2

PRACTICE · 9 MIN

SILVER

Write each number in index form

1. 72
2. 100
3. 84
4. 225
5. 360

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $72 = \mathbf{2^3 \times 3^2}$

2. $100 = \mathbf{2^2 \times 5^2}$

3. $84 = \mathbf{2^2 \times 3 \times 7}$

4. $225 = \mathbf{3^2 \times 5^2}$

5. $360 = \mathbf{2^3 \times 3^2 \times 5}$

Examples 5 and 6 — Using Prime Factors

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Example 5 (2 marks)

Write 144 as a product of prime factors.

Use your answer to find $\sqrt{144}$.

Example 6 (2 marks)

$n = 2^3 \times 5 \times 7$

Find the value of n.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$144 = 2^4 \times 3^2$

$\sqrt{144} = 2^2 \times 3$

$= \mathbf{12}$

Tip: halve each power for a square root.

Example 6

$n = 8 \times 5 \times 7$

$\mathbf{n = 280}$

Tip: work out the powers first.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

1. Write 400 as a product of prime factors in index form. (2 marks)

2. Find the value of $2^2 \times 3^3 \times 5$. (1 mark)

3. Write 324 in prime factors. Use it to find $\sqrt{324}$. (3 marks)

4. What is the smallest number that 18 must be multiplied by to make a square number? (2 marks)

5. Write 1050 as a product of prime factors. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $400 = \mathbf{2^4 \times 5^2}$

2. $4 \times 27 \times 5 = \mathbf{540}$

3. $324 = 2^2 \times 3^4, \quad \sqrt{324} = 2 \times 3^2 = \mathbf{18}$

4. $18 = 2 \times 3^2$, **multiply by 2**

5. $1050 = \mathbf{2 \times 3 \times 5^2 \times 7}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write 28 as a product of prime factors

SILVER Write 150 in index form

GOLD Find the value of $2^4 \times 3 \times 5$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $28 = \mathbf{2^2 \times 7}$

SILVER $150 = \mathbf{2 \times 3 \times 5^2}$

GOLD $16 \times 3 \times 5 = \mathbf{240}$

Exam Practice — Prime Factors

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (2 marks)

Here is a list of numbers.

15 17 21 23 27 29

Write down all the prime numbers in the list.

.....

Question 2 (2 marks)

Write 60 as a product of its prime factors.

.....

Exam Practice — Index Form

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3 (2 marks)

Write 168 as a product of its prime factors.

Give your answer in index form.

.....

Question 4 (2 marks)

Write 252 as a product of its prime factors.

Give your answer in index form.

.....

Exam Practice — Using Prime Factors

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: answer on the line, show all your working

Question 5

(3 marks)

(a) Write 540 as a product of its prime factors in index form. (2)

.....

(b) Find the smallest whole number that 540 must be multiplied by to give a square number. (1)

.....

Question 6

(2 marks)

$n = 2^4 \times 3 \times 5^2$

Write n^2 as a product of its prime factors in index form.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

17, 23, 29 — B2

B1: two right, none wrong

Tip: 21 = 3 × 7 and 27 = 3 × 9.

Question 2

60 = 6 × 10 — M1

= 2 × 3 × 2 × 5

= 2 × 2 × 3 × 5 — A1

Tip: keep splitting until every branch is prime.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$168 = 2 \times 84 = 2 \times 2 \times 42$ **M1**

$= 2 \times 2 \times 2 \times 3 \times 7$

$= 2^3 \times 3 \times 7$ **A1**

Tip: count each prime.

Question 4

$252 = 4 \times 63 = 2 \times 2 \times 9 \times 7$ **M1**

$= 2^2 \times 3^2 \times 7$ **A1**

Tip: write the primes in order.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) $540 = 4 \times 135 = 4 \times 5 \times 27$ — M1

$= 2^2 \times 3^3 \times 5$ — A1

(b) need 3 and 5 once more

$3 \times 5 = 15$ — B1

Tip: a square number has even powers.

Question 6

double every power — M1

$n^2 = 2^8 \times 3^2 \times 5^4$ — A1

Tip: $(2^4)^2 = 2^8$.